

Aggregating Ethics: Moral Uncertainty Aggregation for LLM Alignment

MSc Dissertation Draft

1 Introduction

As AI systems become increasingly capable and influential, the problem of ensuring they behave in accordance with human values grows correspondingly urgent. The alignment problem [6] is the task of ensuring that an AI system pursues the goals and honours the values its developers and users intend, rather than proxies that merely correlate with them. In practice, the dominant approach to aligning large language models is Reinforcement Learning from Human Feedback (RLHF) [29], in which human raters compare model outputs and their preferences are distilled into a reward signal that guides fine-tuning.

These raters often disagree with one another, particularly in normatively ambiguous moral dilemmas, where there is no universal consensus on right or wrong. Crucially, these disagreements do not stem merely from personal idiosyncrasy or noise. They reflect principled differences between competing ethical frameworks. A utilitarian, a Kantian, and an Ubuntu thinker may each reach a different conclusion while reasoning coherently from within their own tradition. In RLHF, this disagreement is not preserved, however. A single reward model is trained to fit the aggregate of many annotators' conflicting preferences, converging on a signal that reflects the pool's majority tendency and offers no explicit account of how competing moral views ought to be combined [5].

This dissertation investigates whether the choice of moral aggregation method produces measurably different LLM behaviour when applied to normatively ambiguous dilemmas. Drawing on MacAskill et al.'s formal framework for decision-making under moral uncertainty [20], I implement three aggregation schemes (Expected Choiceworthiness, Maximin, and Nash Moral Parliament) as competing response-selection criteria for a language model, alongside a single-theory utilitarian baseline. Candidate responses are scored by three ethical reward models operationalising three distinct moral traditions, namely utilitarian, deontological, and Ubuntu. I test whether different aggregation methods select systematically different responses, and if so, where and why the methods diverge. This question has growing practical importance, and as users increasingly turn to language models for guidance on sensitive personal and ethical decisions, the implicit moral character of these systems is no longer a theoretical concern. It is a design choice with real consequences.

Stated precisely, the question is this: *holding fixed a pool of candidate responses to a morally contested dilemma, and holding fixed a set of validated judges representing utilitarian, deontological, and Ubuntu evaluation, does the choice of*

aggregation rule change which response is selected, and does it change the action that response recommends? The two halves are deliberately distinct. Two rules may prefer different texts that steer a reader toward the same course of action, which is disagreement over justification. Or, they may prefer texts whose recommendations are opposed, which is disagreement over what ought to be done. The design measures both (Section 4.7).

The dissertation makes three contributions.

1. **An end-to-end pipeline for moral-uncertainty aggregation.** Five hundred everyday dilemmas filtered for cross-framework tension, each with sixteen candidate responses. Three framework judges score all 8,000 responses, and four selection rules are applied to the normalised scores, with Ubuntu included on equal terms with the two Western traditions (Section 4).
2. **A behavioural validation method for framework judges.** Each judge is tested against fifteen operational axioms drawn from the normative literature, with directional predictions for the other two judges and a held-out set guarding against prompt overfitting (Section 4.5). The harness is reusable for certifying any purported framework judge.
3. **A two-level, tie-robust measure of divergence.** Selections are treated as winner sets rather than broken by arbitrary tie-breaks, and disagreement over the chosen text is reported separately from disagreement over the recommended action, so that no reported disagreement could be removed by a choice of tie-break (Section 4.7).

Section 2 sets out the formal problem and the aggregation proposals this study operationalises, Section 3 situates the work in the surrounding literature, and Sections 4 and 5 describe the pipeline and its results.

2 Moral Uncertainty

Moral uncertainty is uncertainty about which moral theory is correct, as distinct from uncertainty about the facts. An agent may know exactly what an action will cause, yet still not know whether outcomes are all that morally matters. Formally, the agent faces options A and entertains rival theories $T_1, \dots, T_n$ with credences $w_1, \dots, w_n$ summing to one, each grading the options with a real-valued choiceworthiness function c_i . The problem is which option is rational to choose, given credences and verdicts together [20]. The sceptical reply, "act on the theory you find most plausible", ignores stakes in the same way that acting on the most probable state

of the world, whatever the payoffs, ignores stakes under empirical uncertainty. A theory held at credence 0.4 that mildly favours an option should not silence theories with combined credence 0.6 that condemn it as catastrophic [20].

The standard constructive proposal, maximising **expected choiceworthiness** (MEC), imports the apparatus of expected utility [19, 20]:

$$a^* = \arg \max_{a \in A} \sum_{i=1}^n w_i c_i(a).$$

MEC presupposes that the c_i are cardinal and expressed in a common unit, though the theories themselves supply neither. There is no theory-neutral answer to how many units of utilitarian welfare a broken Kantian duty is worth. This is the *intertheoretic value problem*, the central obstacle to any additive rule [20]. The usual response is normalisation, which attempts to place the theories on a common scale by statistical fiat. Lockhart [19] proposes equalising each theory’s range over the options, and Cotton-Barratt et al. [8] argue that equalising variance better gives every theory an equal say. A further dispute concerns scope. Do we normalise over the present decision’s options alone, or over all options a theory could ever grade? Both problem and responses recur here, as the judges differ in how much of the 0–10 scale they use, and the global z -scoring of Section 4.6.3 is a broad-scope variance normalisation.

MEC also inherits the pathologies of expected utility, chief among them *fanaticism*. Because credence and stakes are multiplied, a theory held at very low credence can still dominate the sum whenever it declares the stakes extreme enough, and the agent ends up governed by a theory it barely believes [20]. That worry motivates rules that do not add. Bostrom [4] pictures the theories as a **moral parliament**, delegates who hold seats in proportion to credence and negotiate a compromise rather than being summed, a proposal later formalised by Newberry and Ord [27]. The parliament is a picture rather than a rule, and bargaining theory supplies the missing mechanism. The **Nash solution** [26], applied to moral uncertainty by Greaves and Cotton-Barratt [12], selects the option that maximises the product of each theory’s gain over a disagreement point. This rewards compromise, since an option that leaves any one theory near its disagreement point has its product destroyed however well the others fare. Treating the theories as voters instead imports the impossibility results of social choice [1] and, being rank-based, discards the magnitudes MEC trades on. Finally, a risk-averse agent may prefer to protect the worst-served theory rather than maximise an average. This yields **maximin**, which grades each option by the lowest choiceworthiness any theory assigns it, and then selects the option whose worst grade is highest. Where MEC asks which option is best on balance, maximin asks which option the most dissatisfied theory can best live with.

Section 4.6.1 instantiates one member of each family, MEC for the expectational approach, the Nash product for bargaining, maximin for the worst case, and a single-theory **utilitarian baseline** for a system that ignores moral uncertainty altogether. Credences are held equal ($w_i = 1/3$) in the main analysis, so differences between rules come from

the aggregation function rather than from a prior ranking of the traditions. A credence sensitivity analysis later relaxes this assumption (Section 5.5), and Table 3 offers a concrete mathematical summary of the four rules as they are applied in the pipeline.

3 Related Work

RESUME PROOF READING HERE Computational treatments of moral uncertainty are recent and sparse. Ecoffet and Lehman [10] give the first reinforcement-learning implementation, encoding rival theories as reward functions in gridworlds and comparing expectation-style aggregation with voting. They identify intertheoretic comparison as the immediate practical obstacle. Tennant et al. [33] fine-tune language-model agents on intrinsic rewards that encode a utilitarian or a deontological objective, one tradition at a time. More recent work proposes language-model pipelines that operationalise moral-uncertainty frameworks for decision support [30], or studies how uncertainty about values surfaces in human and model judgments [18]. A parallel line aggregates across people rather than theories. Conitzer et al. [7] argue that social choice should govern the aggregation of diverse human feedback in alignment, and the Moral Machine project distilled millions of verdicts on autonomous-vehicle dilemmas into a voting-based decision system [2, 28]. None of this work compares the philosophical aggregation rules as selection criteria over natural-language advice, which is the setting studied here.

The alignment literature has begun to treat value disagreement as a problem in its own right. Gabriel [11] distinguishes the technical problem of fitting a system to given values from the normative problem of deciding whose values those should be. Sorensen et al. [32] argue for *pluralistic alignment*, in which a system represents several reasonable value systems rather than one aggregate. Moral uncertainty is one formally specified way of doing that, with explicit credences and an explicit rule in place of an unexamined average. The inclusion of Ubuntu follows from the same motive. African ethical traditions are nearly absent from alignment research, although Mhlambi [24] argues that Ubuntu’s relational conception of personhood bears directly on AI governance, and Metz’s reconstruction of Ubuntu as a moral theory [22] is precise enough to operationalise. It enters this pipeline as a first-class framework, with credence equal to the two Western traditions.

On the evaluation side, ETHICS tests whether models can classify right and wrong under utilitarian, deontological, virtue, and justice framings [14], and Delphi trains a descriptive model of commonsense moral judgment [15]. Both score isolated actions rather than select among competing responses. The dilemmas used here come from DailyDilemmas, a corpus of everyday quandaries annotated with the values each side implicates [9]; those annotations drive the filter of Section 4.2. Because the judges are language models, they inherit that method’s known biases, verbosity among them [35], and are not taken on trust: the axiom harness of Section 4.5 tests each judge directionally, on pairs length- and register-matched so those biases have little to grip.

The philosophical literature, then, supplies competing rules but no evidence on whether the choice among them changes the behaviour of a language-model system. The machine-ethics literature either aggregates across people or encodes a single tradition, and validates its judges thinly if at all. This dissertation is the missing experiment, with the philosophical rules implemented end to end, the judges tested behaviourally before use, a non-Western tradition included on equal terms, and divergence measured at the level that matters, the recommended action.

4 Methodology

4.1 Overview

The pipeline has four stages. First, 500 dilemmas are drawn from DailyDilemmas and filtered toward cases where the frameworks are predicted to disagree (Section 4.2). Second, a local `llama3.1:8b` model produces sixteen candidate responses per dilemma under a single neutral prompt, so diversity comes from sampling rather than framework-flavoured instructions (Section 4.3). Third, three prompted `gpt-4o-mini` judges score every response on a 0–10 scale, after an axiom harness has checked that each discriminates in the direction its theory requires (Section 4.5). Fourth, the four selection rules are applied to the globally *z*-scored ratings, with disagreement measured over the chosen text and the recommended action (Sections 4.6 and 4.7). Credences are equal throughout, and only the aggregation function varies.

4.2 Dilemma dataset

DailyDilemmas is a corpus of everyday moral quandaries, each posed as a binary choice between a named `TODO` action and a named `NOT_TODO` action, and annotated with the values each side implicates [9]. Those annotations are a keyword proxy, not a judgment by the later ethical judges. Each annotated value is mapped onto utilitarian, deontological, or Ubuntu keyword sets. Matches are counted on each side, and a dilemma is kept only when the three frameworks split, with at least one preferring `TODO` and at least one preferring `NOT_TODO`. The kept set is ranked by how evenly the frameworks pull (*balance*) and by how much annotated evidence backs the leans (*confidence*), and the top 500 are retained.

The filter is a candidate-selection heuristic. Over half of the distinct values in the source never match a keyword and are dropped, substring matching misclassifies some antonyms, and the mapping itself is a judgment. The 500 dilemmas are therefore a pool expected to be contested, not a set on which the judges are known to disagree. Whether they do is settled only by the scores, and part of the observed agreement is an artefact of the filter admitting dilemmas that proved less torn than the annotations suggested (Section 5.6).

4.3 Response Generation

Sixteen candidate responses per dilemma are sampled from a local `llama3.1:8b` model with no framework-flavoured

system prompts. Priming the generator with a moral framework would manufacture the very disagreement the experiment sets out to detect, so the frameworks enter only at the judging stage and diversity comes from decoding alone: temperature 1.1 with nucleus sampling at $p = 0.95$. The temperature sits deliberately above the conventional range, because the rules can only be distinguished by a pool that spreads across the options, and a conservative setting would return sixteen near-identical answers. The nucleus cap keeps that heat from degenerating into incoherence. Generation is therefore hot while scoring is cold (temperature 0.1, Section 4.5), widening the pool while keeping the measuring instrument stable, and the homogeneity that nonetheless remains (Section 5.6) is a finding about the generator rather than an artefact of cautious decoding.

A recurring artefact of this stage is refusal. On a subset of sensitive dilemmas (abortion, sexual addiction, wishing harm on another person), a substantial fraction of the sixteen candidates decline to recommend an action, typically with a short “I can’t...” disclaimer. These refusals are left in the pool and scored like any other response, and they are evidence that safety alignment in the *generator* can shrink the effective action space before any judge or aggregation rule is applied.

4.4 Reward Models

The three judges are the same underlying model, `gpt-4o-mini`, steered by a system prompt. Each prompt states the theory’s criterion of rightness, supplies anchors for the full 0–10 range, and asks for a single integer and nothing else. The utilitarian prompt scores by total wellbeing and expected outcomes, and the deontological prompt by respect for persons, perfect duties, and the prohibition on using someone merely as a means. The Ubuntu prompt scores by harmony, solidarity, and the quality of the relationships a response would leave behind. The same model is used for all three so that differences in score can be attributed to the prompt rather than to differences in base capability.

4.5 Judge Validation via Axiom Testing

A prompt is not a proof of fidelity. Before the scores can be trusted as a proxy for utilitarian, deontological, or Ubuntu evaluation, it must be shown that each judge discriminates between responses in the direction its assigned theory predicts. Quantifying *how* utilitarian a line of reasoning is has no agreed metric, so I adopt a *differential* test. Each theory is operationalised as a small set of defining axioms drawn from the normative-ethics literature, and the judge must prefer the response that better satisfies the relevant axiom.

4.5.1 Operationalising the theories as axioms

Each framework is decomposed into five operational axioms, each capturing one central commitment of the theory. These are not offered as canonical axiomatisations (none of these traditions comes as a settled list of propositions) but as a defensible operationalisation, grounded in the literature, of

the commitments each judge is meant to track. Table 1 summarises all fifteen, with the commitment each targets and the cross-framework contrasts they predict.

Utilitarianism. Classical utilitarianism [3, 25] holds that the right action is whichever produces the most good overall. The independent commitments that structure this claim, articulated most rigorously by Sidgwick [31], supply the five axioms Ut1–Ut5, namely *aggregation* (welfare is summed across all affected persons, so helping more people is better), *consequentialism* (only outcomes bear on rightness, not processes), *maximisation under uncertainty* (expected welfare governs risky choices), the absence of *side-constraints* (sacrificing one to benefit many is permitted when it maximises the total), and *welfarism* (only wellbeing is intrinsically valuable).¹

Deontology. Kantian deontology [16] judges an action by whether its underlying maxim conforms to the moral law, expressed in the *Formula of Universal Law* (could the maxim be consistently willed as universal? 4:421)² and the *Formula of Humanity* (persons must always be treated as ends, never merely as means, 4:429), with *perfect duties* (strict prohibitions, such as on false promises) distinguished from *imperfect* ones. Axioms De1–De5 target respect for rational agency, the means–ends prohibition, the priority of duty over consequences, universalisability, and the perfect duty of honesty, the last drawing on Kant’s argument that truthfulness is unconditional even where a lie would prevent harm [17].

Ubuntu. Ubuntu locates moral value in the quality of relationships between people; Mbiti’s “I am because we are, and since we are, therefore I am” [21, p.106] makes personhood constitutively communal. Metz reconstructs the tradition as a moral theory, “an action is right just insofar as it produces harmony and reduces discord” [22, p.338], where harmony combines *identity*, the sharing of a way of life, with *solidarity*, acting for others’ good out of genuine good will [23]. Gyekye’s moderate communitarianism keeps individual agency exercised within communal life [13], and Tutu [34] extends the tradition to justice, prioritising the restoration of harmony over retribution. Ubuntu earns its place in this study because it generates genuine disagreement with both Western frameworks wherever they treat the people affected as unconnected individuals. Axioms Ub1–Ub5 target solidarity, relational awareness, restorative justice, communal harmony, and relational personhood.

4.5.2 Test construction

For each axiom I author three dilemmas (Ut1a, Ut1b, Ut1c, ..., Ub5c), 45 in total. Every dilemma presents a pair of

candidate responses constructed to differ *only* along the dimension the axiom isolates, with two confounds controlled explicitly. Pairs are *length-matched* (character-count ratio < 1.3 , median 1.05) against verbosity bias, and *register-matched* (parallel syntax, no one-sided hedging or filler) against a fluency bias that rewards the more polished response irrespective of content.

Each dilemma carries a predicted direction of preference ($a > b$, $b > a$, or null for no prediction) for *all three* judges, not only the judge whose framework the axiom belongs to. In the communal-harmony dilemma Ub4b, for instance, a company can move its meetings online or keep gathering in person. The utilitarian judge is predicted to prefer the efficiency of moving online, the Ubuntu judge to prefer the response that keeps the ties of face-to-face contact, and the deontological judge, indifferent to the dimension varied, carries a null. The framework-preferred answer alternates between the A and B slots across dilemmas.

4.5.3 Scoring and pass criterion

Crucially, the axiom responses are graded by the *same* judge functions used in the main pipeline. Each response is presented to the judge *together with the dilemma it addresses* and scored independently on a 0–10 scale. The judge never sees the two candidate responses as a pair (so there is no presentation order to counterbalance), but it does see the scenario that gives a response its meaning. A predicted cell *passes* when the sign of the score difference matches the prediction, and an exact tie fails a directional prediction. A tolerance parameter (set to 0 here) can discount one-point scoring noise if raised. Judges score at temperature 0.1, and owing to the mild stochasticity the full harness was run five times, with results reported as averages over those runs.

4.5.4 Results

After the prompt iteration detailed in Section 4.5.5, the results form a 3×3 grid of judges against axiom frameworks (Table 2). The **diagonal is the validation target**: each judge on its own framework’s axioms. Off-diagonal cells encode predictions about where the frameworks should *disagree*, so a failed off-diagonal cell is a failed prediction about cross-framework contrast, not evidence that a judge misapplies its own theory.

Across the 75 cells carrying a non-null prediction, the judges matched the predicted direction on 86.9% on average (per run: 88, 87, 87, 85, 88%). On the diagonal they are near-perfect, at a mean of 99.1%; the deontological and Ubuntu judges pass all fifteen of their own cells in every run.

Table 2: Mean pass rates over five runs (predicted cells in parentheses). Diagonal entries in bold, and — marks a null prediction.

Judge	Ut axioms	De axioms	Ub axioms
Utilitarian	97% (15)	63% (12)	67% (3)
Deontological	100% (6)	100% (15)	33% (6)
Ubuntu	—	100% (3)	100% (15)

¹Impartiality receives no axiom of its own. If one option produces more welfare, Ut1 already tests it, and if two options produce equal welfare and differ only in *who* is helped, impartiality counts them as tied and gives no direction to test. It is covered indirectly through the equal weighting of persons in Ut1 and Ut5.

²References to Kant give the Academy-edition volume and page.

Table 1: Fifteen validation axioms, grouped by framework. Each axiom states the property the higher-scoring response should satisfy. Cross-framework contrast indicates where other judges are predicted to disagree. The diagonal column reports own-framework pass rate across the three dilemmas per axiom in a representative run; five-run means are reported in the text, and Ut3 is marginal (see Section 4.5.4).

ID	Axiom (higher-scoring response should...)	Commitment	Cross-fw contrast	con-	Diag.
<i>Utilitarianism</i>					
Ut1	help more people	Aggregation (sum-ranking)	none		3/3
Ut2	secure the better outcome even through worse process	Consequentialism	Deont.		3/3
Ut3	maximise <i>expected</i> wellbeing, even against a certain option	Maximisation under uncertainty	none		2/3
Ut4	sacrifice one to benefit many	No side-constraints on sacrifice	Deont.		3/3
Ut5	treat only wellbeing as intrinsically valuable	Welfarism	none		3/3
<i>Deontology</i>					
De1	respect a competent person’s decision rather than override it for their own good	Respect for rational agency	Util.		3/3
De2	refuse to use a person merely as a means, despite a good outcome	Formula of Humanity (means–ends)	Util.		3/3
De3	keep a binding duty rather than break it for better consequences	Priority of duty over consequences	Util.		3/3
De4	refuse an exception for oneself or favoured persons that could not be willed universally	Formula of Universal Law	none		3/3
De5	tell the truth rather than deceive beneficially	Perfect duty of honesty	Util.		3/3
<i>Ubuntu</i>					
Ub1	act from concern for another’s good, with outcomes held equal	Solidarity	none		3/3
Ub2	weigh impact on relationships, not just practicality	Relational impact	none		3/3
Ub3	restore harmony rather than assign blame and punishment	Restorative over retributive	Deont.		3/3
Ub4	strengthen communal bonds rather than weaken them	Communal harmony	Util.		3/3
Ub5	recognise a person as relationally embedded, not atomistic	Relational personhood	Deont.		3/3

The only diagonal instability is Ut3a, a risk-neutrality case pitting a one-in-five chance of saving 100 lives (expected value 20) against saving 5 for certain; the utilitarian judge passes in three of five runs, scoring the options within a point of one another. This reflects a residual, alignment-instilled caution toward gambles that will probably yield nothing rather than an inability to reason as a utilitarian: the same judge takes the gamble decisively when the upside is very large (Ut3c, a one-in-ten chance of helping 100,000), so it weighs expected value but discounts risky options near indifference.

The remaining misses are all off-diagonal and, aside from De2a under the utilitarian judge (3/5), stable across runs. They fall into two classes: philosophically contestable predictions (the deontological judge preferred restorative to retributive responses on all three Ub3 dilemmas; the utilitarian judge missed two of three De1 respect-for-agency cells) and near-ties, where both responses score within a point in every run so no direction emerges (Ub4c; Ub5b under the deontological judge). The weakest entry in Table 2, the deontological judge on Ubuntu axioms (33%), is exactly these two classes alongside a perfect diagonal.

One substantive finding deserves emphasis. On the honesty axiom (De5), the utilitarian judge failed to reward welfare-improving deception on two of three dilemmas, scor-

ing both options low in all five runs, against the strict-welfarist prediction. Alignment training appears to instil an honesty guardrail that persists even when the model is instructed to reason as a strict consequentialist, a limitation worth bearing in mind whenever aligned LLMs are used as ethical evaluators.

4.5.5 Held-out validation

Two judge prompts were revised in response to observed axiom behaviour, which raises a circularity concern: a clause added after seeing a failure might fit those dilemmas rather than reflect a general capability. The deontological judge gained a clause making truthfulness a perfect duty that outranks a better outcome secured by lying; the utilitarian judge lost a clause (“a larger but unlikely benefit does not automatically outrank a smaller but more certain one”) that encoded risk-*aversion* rather than classical risk-neutrality, a theory-faithfulness correction that improved Ut3 from a reliable miss to the marginal pass above.

To test for overfitting, I froze the prompts, authored six fresh dilemmas of the same construction (three honesty, three risk-neutrality), and ran five times. The honesty clause generalised cleanly, all three new dilemmas passing in every run; the risk-neutrality guidance generalised only direction-

ally, the three new cases passing in five, three, and four runs with scores again compressed near indifference. Held-out performance overall was 90%: the honesty edit reflects a genuine disposition, the risk edit correct-but-marginal behaviour consistent with the residual risk-aversion above.

Iteration and limitations. Two further prompt improvements were axiom-independent: vague “be harsh” instructions became the anchored 0–10 rubrics after early runs showed severe score compression, and responses are scored in the context of their dilemma rather than in isolation. Caveats remain. Each axiom rests on only three dilemmas, and the harness validates *directional discrimination on constructed, deliberately clean cases*, not correct reasoning on the messier dilemmas of the main study. Cross-framework coverage is also uneven: the Ubuntu judge is predicted to contrast with the others on only one non-Ubuntu axiom (De2), so the harness gives little evidence that it *diverges* from the other judges as the theory predicts, leaving open the possibility that it is a generically pro-communal scorer rather than a distinctively Ubuntu one.

4.6 Aggregation

4.6.1 The aggregation rules

For a dilemma with candidate responses R , let $c_i(R)$ be the choiceworthiness of R under framework $i \in \{\text{Ut}, \text{De}, \text{Ub}\}$, with equal credences $w_i = 1/3$ in the main analysis so that differences between rules come from the aggregation function rather than from a prior ranking of the traditions (a sensitivity analysis relaxes this; Section 5.5). Each rule selects

$$R^* = \arg \max_R f(c_1(R), c_2(R), c_3(R)),$$

so the four rules differ only in f (Table 3); what is substituted for c_i is settled in Sections 4.6.2 and 4.6.3. The **baseline** consults the utilitarian judge alone, standing in for a system aligned to one tradition and giving the comparison point for what aggregation buys. **Expected choiceworthiness** averages the three scores [20], so a large enough gain for one theory can outvote the others. **Maximin** selects the response whose worst-treated framework fares best, refusing the trade EC allows; it depends on levels, not only rankings. The **Nash moral parliament** maximises the product of each framework’s surplus S_i above a disagreement point [26], rewarding balance: a response scoring near zero for any framework has its product destroyed however well the others rate it. (The weighted Nash solution $\prod_i S_i^{w_i}$ reduces to the plain product under equal credences, since a common power preserves the ordering.) Raw-score ties are common and survive normalisation; the convention for handling them is given in Section 4.7.2.

4.6.2 The intertheoretic value problem

Raw 0–10 scores cannot be aggregated as if they were already comparable. A judge that spreads its scores widely (say 2–8) will swing EC, Maximin, and Nash more than a judge stuck in

Table 3: The four selection rules. c_i is the normalised choiceworthiness under framework i , and S_i the Nash surplus defined in Section 4.6.3.

Rule	Objective f	Character
Baseline	c_{Ut}	Single tradition; disagreement ignored
EC	$\sum_i w_i c_i$	Averages; permits trade-offs
Maximin	$\min_i c_i$	Protects the worst-served framework
Nash	$\prod_i S_i$	Rewards balance; punishes near-zero scores

a narrow band (7–9), even under equal credences. This is the **intertheoretic value problem** of Section 2 in statistical form [20]: plugging unnormalised scores into a shared formula lets scale artefacts, not only moral content, decide the outcome. The axiom harness never faces this, since it checks ordering *within* a single judge, but normalisation is required once judges are combined. Scores are therefore normalised so that each framework influences the choice by how it ranks the candidates, not by how much of the scale it happens to use.

4.6.3 Global z-scoring

Each judge j ’s pooled raw scores over every dilemma and candidate yield a mean μ_j and standard deviation σ_j , and each raw score x is replaced with $z = (x - \mu_j)/\sigma_j$. Every framework is then centred at zero with unit spread, so a one-point lead no longer counts for more under a wide-ranging judge than under a compressed one.

The scope is deliberately *global* rather than per-dilemma. Per-dilemma z -scoring divides by the standard deviation of that dilemma’s sixteen candidates alone; when a judge scores them all within a narrow band that deviation is tiny, a one-point gap is stretched into a large one, and a judge that rated every candidate between 8 and 9 is reported as decisively preferring one of them. Global μ_j and σ_j measure each gap against how much the judge varies across the whole corpus, so genuine near-indifference stays small.

EC and Maximin use these z -scores directly. Nash cannot, since products of negative values are not meaningful as choiceworthiness; it needs non-negative gains above a disagreement point, fixed a priori at a raw score of 0 (absence of moral worth on the rubric), not at the worst score observed in the sample. Subtracting the z -score of raw 0 from each response’s z -score simplifies to the surplus $S_i = X_i/\sigma_i$, where X_i is the raw score under judge i . Because the Nash product is scale-invariant, dividing by σ_i does not change which response is selected relative to a raw product of the X_i ; the substantive choice is the disagreement point at raw 0.

4.7 Divergence Measurement

Divergence is not a single quantity. Two rules may select different response texts that direct the reader toward the same course of action, which is disagreement over justification and rhetoric, or they may select texts whose core recommenda-

tions are opposed, which is disagreement over what ought to be done. The first reveals how the choice of rule shapes the reasoning a system presents; the second is the level at which the research question is posed. Both are measured and reported separately.

4.7.1 Recommendation labelling

The source dataset records, for every dilemma, a pair of canonical actions: a `TO_DO` action and a `NOT_TO_DO` action. Each of the sixteen candidate responses is assigned one of four labels according to the action it endorses:

- `TO_DO`: the response points the reader toward the named to-do action.
- `NOT_TO_DO`: it points toward the named not-to-do action.
- `REFUSAL`: it declines to engage with the dilemma at all (typically a short “I cannot advise on this” disclaimer, with no recommendation attached).
- `OTHER`: a distinct third course, or no discernible lean toward either named action.

The threshold for the two action labels is deliberately permissive: a response that recommends an action tentatively, conditionally, or in softened form is still recorded as endorsing it, since the question is which way the text would move a reader, not how forcefully it argues. An earlier revision of the prompt reserved the action labels for confident endorsements and routed cautious answers into a residual “hedge” category, which conflated hesitancy with refusal and split near-identical responses across categories; `REFUSAL` is accordingly restricted to responses that offer no recommendation of any kind.

Labels are assigned *post hoc* by a separate model (`gpt-4o-mini`) rather than emitted by the generator alongside each response: a self-reported label is a claim about intent rather than an independent reading of the text, and requiring the generator to commit to a label while writing risks the label shaping the prose. Post-hoc annotation keeps the labels independent of generation at the cost of a second source of error, so each label is stored with the short phrase in the response that carries the recommendation, allowing the labelling to be audited without re-reading the corpus.

Two residual weaknesses are acknowledged at the outset. `OTHER` is a residual category, so a shared `OTHER` is weaker evidence of agreement than a shared action label (two responses may endorse quite different third courses); after labelling it proves small, at 63 responses (Section 5.1). And the labels have not been validated against human ground truth: automatic consistency checks are reported in Section 5.1, and a hand-labelled gold sample is left to future work.

4.7.2 Set-valued winners

On a 0–10 integer rubric, distinct responses frequently receive identical score patterns, and because global z -scoring is

an affine transform of each judge’s raw scores, such ties survive normalisation intact. Several candidates may therefore share the top value under a given rule. Rather than breaking ties arbitrarily, each rule’s selection is treated as a *winner set*: all candidates whose value lies within 10^{-9} of the maximum, a tolerance that only absorbs floating-point noise and is orders of magnitude smaller than any genuine gap in the data. A random tie-break would have made the results depend on an arbitrary seed; under the set-valued convention, every reported disagreement is one that no choice of tie-break could remove.

4.7.3 Agreement at two levels

Let S_A and S_B denote the winner sets of rules A and B on a given dilemma, and let $\ell(\cdot)$ denote the recommendation label of a response.

Response level. Rules A and B can agree on a response when their winner sets overlap, $S_A \cap S_B \neq \emptyset$, and diverge when the sets are disjoint. In the disjoint case no tie-breaking convention, however favourable, could produce a common winner, so the disjoint count is a lower bound on disagreement. A corresponding upper bound, used only as a robustness check, is the probability that two uniform draws from S_A and S_B select different responses, $1 - |S_A \cap S_B| / (|S_A| \cdot |S_B|)$.

Recommendation level. Rules A and B can agree on an action when some label appears in both winner sets, $\{\ell(r) : r \in S_A\} \cap \{\ell(r) : r \in S_B\} \neq \emptyset$, and diverge when the label sets share nothing, as when one winner set is entirely `TO_DO` and the other entirely `NOT_TO_DO`. This criterion answers whether the rules would direct a user toward different courses of action.

Both definitions extend to all four rules at once: the rules are unanimous at a level when the intersection over all four winner sets (respectively, label sets) is non-empty. Response-level unanimity entails recommendation-level unanimity, since a shared response carries a shared label; the converse fails, and the gap between the two is the quantity of interest in Section 5.3. Two kinds of nominal agreement are not treated as agreement on a moral position and are counted separately: a shared `REFUSAL` records only that the generator declined to answer, and a shared `OTHER` only that both winners depart from the binary, not that they depart in the same direction.

5 Results

The pipeline is applied to all 500 dilemmas: sixteen candidate responses per dilemma, scored by the three judges, normalised as in Section 4.6.3, and ranked by the four rules of Table 3. The results are presented in five stages: the composition of the labelled pool, the prevalence of ties, agreement at the response level, agreement at the recommendation level, and the sensitivity of that agreement to the credences.

5.1 The labelled pool

Of the 8,000 candidate responses, 3,195 are labelled `TO_DO`, 4,074 `NOT_TO_DO`, 668 `REFUSAL`, and 63 `OTHER`. Two features of this distribution bear on what follows. Refusals are substantial and structurally distinct: the median refusal runs to 25 words against 308 for other responses, and all but one of the 668 open with an explicit decline. This is the generator-side safety behaviour noted in Section 4.3, and the aggregation rules do sometimes select such responses (refusals occupy forty-six winning slots across the four rules), so a safety refusal can outrank every substantive recommendation available for a dilemma. `OTHER`, by contrast, is rare (0.8%), a genuine residual rather than a repository for cautious answers, which limits the exposure identified in Section 4.7.1.

Because the labels were produced by a second language model, two automatic consistency checks were run. Of the 7,659 labels carrying a supporting quote, 99.1% of the quotes appear verbatim in the response they were drawn from once punctuation and markdown are normalised, the remainder being light paraphrases, which is evidence that the labeller reads the text rather than inventing support. Responses that open with a refusal phrase yet received an action label prove, on inspection, to follow the “disclaimer, then advice” pattern the prompt instructs the labeller to judge on the advice. These checks establish internal consistency, not accuracy: recommendation-level figures carry that caveat until a hand-labelled sample exists.

5.2 Ties

Ties at the top are pervasive and markedly uneven across methods (Table 4). The utilitarian baseline is indifferent among two or more candidates on 383 of 500 dilemmas (77%), with a mean winner set of seven when it ties and 26 dilemmas where its winner set is the entire pool. EC and Nash tie on roughly a third of dilemmas, Maximin on just over half.

The pattern reflects the scoring scale more than moral indifference. Across the sixteen candidates for a dilemma, a judge draws on only 3.6 distinct utilitarian values on average (3.6 deontological, 4.0 Ubuntu), and the candidates realise a mean of 9.4 distinct score triples, so 60% of responses share their exact triple with a competitor for the same dilemma; identical triples remain tied after z -scoring, which is affine. The distributions are also top-heavy (the deontological judge places 58% of its scores in the 8–10 band, the other judges roughly 41%), the ceiling effect anticipated during judge design. The baseline inherits this coarseness directly, consulting one coordinate, whereas the aggregation rules combine three and so resolve some ties the baseline cannot. The practical consequence is the convention of Section 4.7.2: disagreement is computed on winner sets, since a large baseline tie-set that happens not to contain the EC winner is response-level divergence even when every text in both sets recommends the same action. Exactly this possibility motivates the two-level analysis.

Table 4: How often each rule is indifferent among two or more top-scoring responses ($n = 500$ dilemmas). “Complete” means the winner set is all sixteen candidates.

Rule	Ties	Rate	Mean size	Complete
EC	166	33%	3.3	0
Maximin	280	56%	4.9	6
Nash	180	36%	3.3	0
Baseline	383	77%	7.1	26

5.3 Two levels of agreement

Table 5 presents the central result. On 339 of 500 dilemmas (68%), some single response lies in every rule’s winner set, so the four rules can agree on both the text and the action. On a further 105 (21%) no common text exists, yet every winner set still contains the same core recommendation. Only on the remaining 56 (11%) do the rules diverge on the action itself.

Table 5: Agreement of all four rules, at two levels ($n = 500$). Sharing a response implies sharing its label, so the missing cell is empty by construction.

	Count	Share
Same response and same recommendation	339	67.8%
Different response, same recommendation	105	21.0%
Different recommendation	56	11.2%

The figure of 56 answers the research question as it concerns *what to do*, and it is a lower bound: a dilemma counts as action-level divergence only when the four label sets are entirely disjoint. Of the 56, 44 present a direct `TO_DO` versus `NOT_TO_DO` split, five involve a refusal, and the remainder involve `OTHER`.

The 105 dilemmas in the middle row are those where the rules prefer different texts that point the same way. In a minority the difference is a genuine difference of justification, as when one winner counsels “confess, but carefully” where another counsels simply telling the truth, both `TO_DO`. In rather more, the winner sets are non-overlapping subsets of near-paraphrases, typically because the baseline is indifferent among a large collection of similar answers from which the aggregation rules select a different subset. Twenty-four of the 105 are unique-winner splits, the strongest evidence that the choice of rule can change which justification is presented even when the action is unchanged. The raw response-level figure of 161/500 (32%) therefore overstates practical disagreement considerably: most of it is rhetorical, and some of the rhetoric is paraphrase.

Of the 444 recommendation-level agreements, 221 fall on `NOT_TO_DO`, 212 on `TO_DO`, nine on `REFUSAL`, and two on `OTHER`. The nine refusal-agreements record only that the generator declined to answer; they are retained in the 444 but excluded from any claim that the rules share a moral position.

5.4 The structure of the disagreement

Table 6 reports every pair of rules at both levels, and the pattern is consistent. EC and Nash almost never select different

actions (4 of 500 dilemmas, 0.8%); Maximin departs from either more often but still in low single digits; and nearly all action-level disagreement lies between the three uncertainty rules and the utilitarian baseline, a contrast that is itself modest at 38–44 dilemmas per pair (roughly 8%). Consistent with this, of the 56 dilemmas on which the four rules cannot share a recommendation, 31 have the three uncertainty rules agreeing on an action with the baseline apart; the remaining 25 are splits among the uncertainty rules themselves.

Table 6: Pairwise divergence ($n = 500$). Response-level: winner sets are disjoint. Recommendation-level: label sets are disjoint.

Pair	Response	Recommendation
EC vs Nash	7.4%	0.8%
Maximin vs Nash	11.0%	3.8%
EC vs Maximin	20.6%	5.0%
EC vs Baseline	16.6%	7.8%
Nash vs Baseline	21.4%	7.6%
Maximin vs Baseline	26.6%	8.8%

The forced-choice figures are considerably larger (around 70–80% against the baseline, 29% even for EC against Nash) because a large tie-set makes a uniform draw unlikely to land on the one shared candidate. They are reported only as an upper bound: that randomness is no part of the rules, and treating it as a headline would attribute to them a disagreement that belongs to the tie-break.

5.5 Credence sensitivity

To test whether the observed agreement depends on the equal-credence choice, the aggregation stage was re-run twice with a 2:1:1 tilt: credences (0.25, 0.5, 0.25) favouring deontology, and (0.25, 0.25, 0.5) favouring Ubuntu. Only EC and Nash can respond (EC reweights its sum; Nash becomes the asymmetric product $\prod_i S_i^{w_i}$); Maximin protects the worst-served theory whatever its credence and the baseline consults a single judge, so both are unchanged by construction.

The tilts move the rules considerably at the level of text: the deontological tilt hands EC a different winner set on 85 of 500 dilemmas and Nash on 70 (the Ubuntu tilt: 49 and 55). Almost none of that movement reaches the action (Table 7). Action-level disagreement rises from 56 dilemmas to 61 under either tilt, and most dilemmas that lose their shared response merely prefer a different text pointing the same way. What growth there is concentrates against the credence-free baseline (EC versus baseline, 7.8% to 10.4% under the deontological tilt), while EC versus Nash stays near 1%. Doubling a theory’s credence changes which justification wins far more often than what the system recommends; the headline of Table 5 is not an artefact of the equal-credence choice.

5.6 Interpretation

On the quantity the motivating premise concerns, the recommended action, the observed effect is modest. Once justification is set aside, EC and Nash are close to interchangeable,

Table 7: Agreement of all four rules under three credence settings ($n = 500$). Each tilted run gives the named framework credence 0.5 and the other two 0.25.

	Equal	De 0.5	Ub 0.5
Same response, same recommendation	339	316	323
Different response, same recommendation	105	123	116
Different recommendation	56	61	61

and replacing a utilitarian default with any of the three uncertainty rules changes the selected action in fewer than one dilemma in ten, on a pool already filtered toward moral contestedness.

The result remains informative because it locates *where* the machinery matters: rarely between one uncertainty rule and another, occasionally against a single-tradition default, and more often as a change of justification than of action. Three features of the pipeline help explain why the action-level figure is small, and they belong alongside the result. First, the candidate pool is homogeneous: responses come from an 8B-parameter model under one neutral prompt, so many of the sixteen candidates are close restatements of the same side, and every rule is reduced to ranking stylistic variants of one recommendation. Second, the judges sometimes saturate: where nearly every candidate satisfies the relevant duty or relational obligation, the deontological and Ubuntu scores cluster near the top of the scale, EC, Maximin, and Nash become effectively utilitarian-driven, and agreement there is an artefact of compressed scores. Third, the key-word filter of Section 4.2 predicts where disagreement should arise but cannot guarantee it, so part of the pool proved less contested than the annotations suggested.

None of this licenses substituting the larger response-level count for the 56: that count is real but answers a different question, which text is preferred, and it is inflated by ties among paraphrases. The defensible claim is the narrower one Table 6 supports: modelling moral uncertainty rarely changes the recommended action relative to other uncertainty rules, and changes it relative to a utilitarian default in roughly eight dilemmas in a hundred.

6 Future Work

Four limitations point to concrete extensions.

The measurement scale is coarse, and the tie rates of Section 5.2 are its most visible consequence: a judge that separates sixteen candidates into fewer than four distinct values cannot express fine distinctions, so much of the observed indifference is an artefact of resolution. Re-scoring on a wider rubric is not the simple fix it appears: language-model judges cluster on round numbers, so a 0–100 scale would resolve only rounding ties, not near-paraphrase ties, and the anchors were axiom-validated at their present granularity, so rewriting them yields a new instrument that must be revalidated before any comparison. Meanwhile the set-valued convention keeps the coarseness from inflating the headline.

The recommendation labels require human validation.

The automatic checks establish that the labeller is consistent with its own prompt and its quotes grounded in the text, not that a human would assign the same labels. A stratified gold sample of several hundred responses, labelled blind to the model’s output, would give per-class precision (most importantly on the softened-action boundary) and a prevalence-reweighted estimate of corpus accuracy. Until then, the 56 action-level disagreements are the output of an unvalidated classifier atop an otherwise fully specified pipeline.

A categorical label discards degree. “Gradually distance yourself” and “cut off contact immediately” may share a NOT_TO_DO tag while differing sharply in force. A scalar lean on $[-1, +1]$ recorded alongside the category would let divergence be graded rather than binary, though it should not be folded into the aggregation scores, which measure choiceworthiness, not how firmly a response is phrased.

The homogeneity of the candidate pool is the binding constraint, as Section 5.6 argues. A larger generator, or a procedure designed to produce genuinely opposed courses of action rather than sixteen variants of one side, would give the rules materially more to discriminate between. Were action-level divergence to remain near 11% under such a pool the negative result would be considerably strengthened, and were it to rise the present figure would stand revealed as an artefact of sampling rather than a fact about the rules.

References

- [1] Arrow, K. J. (1951). *Social Choice and Individual Values*. Yale University Press.
- [2] Awad, E., Dsouza, S., Kim, R., Schulz, J., Henrich, J., Shariff, A., Bonnefon, J.-F., & Rahwan, I. (2018). The Moral Machine experiment. *Nature*, 563(7729), 59–64.
- [3] Bentham, J. (1789). *An Introduction to the Principles of Morals and Legislation*. London: T. Payne & Son.
- [4] Bostrom, N. (2009). Moral uncertainty — towards a solution? *Overcoming Bias*.
- [5] Casper, S., et al. (2023). Open problems and fundamental limitations of reinforcement learning from human feedback. *Transactions on Machine Learning Research*.
- [6] Christian, B. (2020). *The Alignment Problem: Machine Learning and Human Values*. W. W. Norton & Company.
- [7] Conitzer, V., et al. (2024). Aggregation problems in machine ethics and AI alignment. *AAAI/ACM Conference on AI, Ethics, and Society*.
- [8] Cotton-Barratt, O., MacAskill, W., & Ord, T. (2020). Statistical normalization methods in interpersonal and intertheoretic comparisons. *Journal of Philosophy*, 117(2), 61–95.
- [9] Chiu, J., et al. (2024). Daily Dilemmas: Revealing value preferences of LLMs with quandaries of daily life. *ICLR 2025*.
- [10] Ecoffet, A. & Lehman, J. (2021). Reinforcement learning under moral uncertainty. *Proceedings of ICML*.
- [11] Gabriel, I. (2020). Artificial intelligence, values, and alignment. *Minds and Machines*, 30(3), 411–437.
- [12] Greaves, H. & Cotton-Barratt, O. (2019). A bargaining-theoretic approach to moral uncertainty. *Global Priorities Institute Working Paper No. 4-2019*, University of Oxford.
- [13] Gyekye, K. (1997). *Tradition and Modernity: Philosophical Reflections on the African Experience*. Oxford University Press.
- [14] Hendrycks, D., et al. (2020). Aligning AI with shared human values. *arXiv:2008.02275*.
- [15] Jiang, L., Hwang, J. D., Bhagavatula, C., Le Bras, R., Forbes, M., Borchardt, J., Liang, J., Etzioni, O., Sap, M., & Choi, Y. (2021). Delphi: Towards machine ethics and norms. *arXiv:2110.07574*.
- [16] Kant, I. (1785). *Groundwork of the Metaphysics of Morals*. (M. Gregor, Trans., 1998). Cambridge University Press.
- [17] Kant, I. (1797). On a supposed right to lie from philanthropic concerns. *Berlinische Blätter*, 1, 301–314.
- [18] Kwon, J., et al. (2025). Dropouts in confidence: Moral uncertainty in human-LLM alignment. *arXiv:2511.13290*.
- [19] Lockhart, T. (2000). *Moral Uncertainty and Its Consequences*. Oxford University Press.
- [20] MacAskill, W., Bykvist, K., & Ord, T. (2020). *Moral Uncertainty*. Oxford University Press.
- [21] Mbiti, J. S. (1969). *African Religions and Philosophy*. Heinemann.
- [22] Metz, T. (2007). Toward an African Moral Theory. *Journal of Political Philosophy*, 15(3), 321–341.
- [23] Metz, T. (2011). Ubuntu as a moral theory and human rights in South Africa. *African Human Rights Law Journal*, 11(2), 532–559.
- [24] Mhlambi, S. (2020). From rationality to relationality: Ubuntu as an ethical and human rights framework for artificial intelligence governance. *Carr Center Discussion Paper*, Harvard Kennedy School.
- [25] Mill, J. S. (1863). *Utilitarianism*. London: Parker, Son & Bourn.
- [26] Nash, J. (1950). The bargaining problem. *Econometrica*, 18(2), 155–162.
- [27] Newberry, T. & Ord, T. (2021). The parliamentary approach to moral uncertainty. *Future of Humanity Institute Technical Report #2021-2*, University of Oxford.
- [28] Noothigattu, R., Gaikwad, S., Awad, E., Dsouza, S., Rahwan, I., Ravikumar, P., & Procaccia, A. D. (2018). A voting-based system for ethical decision making. *Proceedings of AAAI*.
- [29] Ouyang, L., et al. (2022). Training language models to follow instructions with human feedback. *NeurIPS*.
- [30] Rosello, P., et al. (2025). Addressing moral uncertainty using large language models for ethical decision-making. *arXiv:2503.05724*.
- [31] Sidgwick, H. (1907). *The Methods of Ethics* (7th ed.). London: Macmillan.
- [32] Sorensen, T., Moore, J., Fisher, J., Gordon, M., Mireshghallah, N., Rytting, C. M., Ye, A., Jiang, L., Lu, X., Dziri, N., Althoff, T., & Choi, Y. (2024). A roadmap to pluralistic alignment. *Proceedings of ICML*.
- [33] Tennant, E., et al. (2024). Moral alignment for LLM agents. *arXiv:2410.01639*.
- [34] Tutu, D. (1999). *No Future Without Forgiveness*. Rider Books.
- [35] Zheng, L., Chiang, W.-L., Sheng, Y., Zhuang, S., Wu, Z., Zhuang, Y., Lin, Z., Li, Z., Li, D., Xing, E., Zhang, H., Gonzalez, J. E., & Stoica, I. (2023). Judging LLM-as-a-Judge with MT-Bench and Chatbot Arena. *NeurIPS*.